[Model] Unified Finsler Triplet Loss Formulation

Finsler distance, RandomIdentitySampler, batch-hard mining on pid

$$\mathbf{x}_i = (\mathbf{x}_i^{\mathrm{id}}\ \mathbf{x}_i^{\omega})^{\intercal}$$

$$\mathcal{L}_{tri}^{uni} = \frac{1}{B}\sum_{i}^{N}\left[\max_{p:y_p=y_i} d_F(\mathbf{x}_i,\mathbf{x}_p) - \min_{n:y_n\neq y_i} d_F(\mathbf{x}_i,\mathbf{x}_n) + m_{id}\right]_+$$

$$d_F(\mathbf{x},\mathbf{y}) = \|\mathbf{x}_{\mathrm{id}} - \mathbf{y}_{\mathrm{id}}\|_2 + \langle \frac{\mathbf{x}_\omega + \mathbf{y}_\omega}{2}, \mathbf{y}_{id} - \mathbf{x}_{id}\rangle$$

"We want to compare images from different domains, not necessarily same identity"

- Loss can be formulated as a cross-domain constrastive on identity features via Finsler distance
- Encourages the Finsler distance to shrink cross-domain same-identity pairs
    - Asymmetric term must compensate for the domain drift

[Model] Modified Triplet Loss Terms

Motivation: Clean signal separation between identity-related and domain-related loss terms

Idea 1: Split Triplet Loss into Identity-Euclidean + Domain Finsler

$$\mathbf{x}_i = (\mathbf{x}_i^{\text{id}}\ \mathbf{x}_i^{\omega})^{\intercal}$$

1a. Euclidean Identity Triplet: Euclidean distance, RandomIdentitySampler, batch-hard mining on pid

$$\mathcal{L}_{tri}^{id} = \frac{1}{B}\sum_{i}^{N}\left[\max_{p:y_p=y_i} d_E(\mathbf{x}_i^{\text{id}}, \mathbf{x}_p^{\text{id}}) - \min_{n:y_n\neq y_i} d_E(\mathbf{x}_i^{\text{id}}, \mathbf{x}_n^{\text{id}}) + m_{id}\right]_+$$

1b. Finsler Domain Triplet: Finsler distance, RandomDomainSampler, batch-hard mining on domain/camera

$$\mathcal{L}_{tri}^{dom} = \frac{1}{B}\sum_{i}^{B}\left[\max_{p:d_p=d_i} d_F(\mathbf{x}_i, \mathbf{x}_j) - \min_{n:d_n\neq d_i} d_F(\mathbf{x}_i, \mathbf{x}_n) + m_{dom}\right]_+$$

$$d_F(\mathbf{x}, \mathbf{y}) = \|\mathbf{x}_{\text{id}} - \mathbf{y}_{\text{id}}\|_2 + \langle \frac{\mathbf{x}_\omega + \mathbf{y}_\omega}{2}, \mathbf{y}_{id} - \mathbf{x}_{id}\rangle$$

d_i Is the domain/camera label

- Facilitates feature disentanglement and isolates the contributions of asymmetric (Finsler) geometry to domain modelling

Rationale
- Identity invariance stays Euclidean (same-person features close regardless of domain)
- Domain structure is modeled via Finsler (same-domain features clustered in asymmetric space)

Consideration
- We can still fine-tune the margin for Finsler Domain Triplet

Critical analysis
- Domain triplet pulls same-domain embeddings closer and pushes different-domain embeddings apart
- Existing domain loss does the exact opposite; it repels nearest same-domain memory bank neighbors
- If we proceed with domain triplet we will be clustering by domain or camera, which is the failure mode of domain-generalizable person re-identification.
    - Re-introduce camera-style shortcuts
    - Loss learns to cluster arbitrary identities by domain

If we want to learn what is different between domains (drift vector models domain artifacts)
- Positive should be cross-domain same-identity
- Negative should be same-domain same-identity

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1b. (Refined) Cross-domain constrastive loss: Finsler distance, RandomDomainSampler

$$\mathcal{L}_{contrast}^{dom} = \frac{1}{|\mathcal{C}|}\sum_{(i,j)\in\mathcal{C}} d_F(\mathbf{x}_i, \mathbf{x}_j)^2 \qquad \mathcal{C} = \{(i,j) : y_i = y_j, d_i \neq d_j\}$$

$$d_F(\mathbf{x}, \mathbf{y}) = \|\mathbf{x}_{\mathrm{id}} - \mathbf{y}_{\mathrm{id}}\|_2 + \langle \frac{\mathbf{x}_\omega + \mathbf{y}_\omega}{2}, \mathbf{y}_{id} - \mathbf{x}_{id}\rangle$$

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1c. Euclidean L2 Loss on Drift Vector: Euclidean distance, RandomDomainSampler

$$\mathcal{L}_{\text{drift-xcontrast}} = \frac{1}{|\mathcal{C}|}\sum_{(i,j)\in\mathcal{C}} \|\boldsymbol{\omega}_i - \boldsymbol{\omega}_j\|_2^2 \quad \mathcal{C} = \{(i,j) : y_i = y_j,\ c_i \neq c_j\}$$

"We want to compare images from different domains, not necessarily same identity"

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[Model] Modified Triplet Loss Terms -- Results

Short Name	Description	Drift Conditioning	mAP	Rank-1
baseline_1a_only	Euclidean identity triplet only (Idea 1a); no cross-domain contrastive; no domain triplet; no drift-only L2	instance	43.7	43.4
		domain	44.1	45
cam_crossview_contrast	Refined 1b: Finsler distance on same-PID pairs with different global camera id; weight 0.1	instance	42.2	42.2
		domain	42.8	44.1
domain_triplet_no_memory_domain	Batch-hard triplet on camera labels using full Finsler embedding; domain_triplet_weight=1.0; BAU memory domain loss disabled (--no-domain)	instance	24.6	30.4
		domain	24.6	30.4
drift_only_cross_camera_l2	Mean squared L2 on drift vectors for same PID different camera; weight 0.1	instance	43	43.1
		domain	42.8	41.9
domain_triplet_with_memory_domain	Same domain triplet as 3rd but BAU memory domain loss enabled (no --no-domain)	instance	35.5	36.4
		domain	36.2	36.1

[Model] Finslerized ReNorm

short_name	drift_conditioning	mAP	Rank1	eval_epoch
Baseline euclidean	Final Epoch	42.36	41.5	119
	Peak MAP	43.05	42.71	79
unified_finsler	instance			
	domain	42.21	41.79	69
1a + 1b	instance			
	domain	35.55	34.29	69
1a + 1b (refined)	instance	14.76	13.29	59
	domain	32.01	32.14	39
	domain	28.81	29.29	59

This Week

1. [Results] Finslerized ReNorm – Results
2. [Results] Modified Triplet Loss with Euclidean Domain Loss
3. [Results] Asymmetric Ranking during Evaluation
4. [Results] Best model configuration vs. Protocol-2
5. [Theory] Intuition/Justification on why this config improves on Euclidean BAU

Experiments if we want to publish

1. [Sweep] Softmax temperature and domain token loss weight
2. [Ablation] Instance Normalization Layers
3. [Sweep] Omega regularizer scalar weight
4. [Sweep] Scalar multiplier for residual correction term

[Model] Finslerized ReNorm

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	domain	28.81	29.29	59

[Model] Modified Triplet Loss Terms

Short Name	Description	Drift Cond.	Fins. Dom.		Euc. Dom.	
			mAP	R1	mAP	R1
1a	Baseline: Euclidean identity triplet only.	instance	43.6	44.1	**43.9**	**43.9**
		domain	43.7	43.2	43.8	**43.9**
1b	Domain Triplet Only: Finsler triplet mining camera labels	instance	35.1	33.9	30.6	30.6
		domain	33.2	34.1	25.6	27.0
1a + 1b	Conflicting Objectives: Both identity and camera triplets are active simultaneously.	instance	36.2	36.4	36.9	36.2
		domain	36.1	34.9	37.8	38.9
1a + 1b (no L_dom)	Same as above, but disabling BAU domain repulsion to isolate domain-attraction.	instance	36.4	34.6	33.6	35.0
		domain	28.5	24.9	26.4	25.5
1a + Refined 1b	Cross-Camera Contrastive: Shrinks Finsler distance d_F for the *same* identity across *different* cameras	instance	43.7	43.1	43.4	42.8
		domain	43.2	42.1	43.1	42.4
1a + 1c	Drift L_2 Alignment: Explicitly penalizes the L_2 variance of drift vectors for the same identity across cameras.	instance	43.4	43.0	42.9	43.8
		domain	**44.9**	43.8	42.9	43.7
Unified Finsler	Single Finsler Triplet: Replaces the Euclidean identity metric with the full asymmetric Finsler distance d_F.	instance	44.1	44.0	43.4	**43.9**
		domain	44.1	**44.3**	43.4	43.0

Instance Conditioning: RandomIdentitySampler; Domain Conditioning: RandomMultipleGallery

- Orange box shows similar values, which is expected because in both runs, the Finsler injection to BAU is removed such that these runs should be identical.
    - Resnet50_finsler acts as a structurally larger model -> uses a split-head architecture; even when the drift is not explicitly evaluated, the backbone is producing a feature vector of 4096 dimensions
    - Orthgonalization step ensures that the drift branch remains orthogonal to the identity branch; strong regularizer on the convolutional backbone -> forces it to learn a more robust representation
    - Normalization performed *after* slicing the identity dimension -> magnitude distribution before normalization differs

- Green boxes should be identical, since the domain loss is deactivated
    - mAP varies significantly -> arguably due to the conflicting retrieval objectives making the optimization unstable, distinct local minimas?

[Model] Modified Triplet Loss Terms - Analysis

1. **Explicitly Camera Clustering (1b only)**: Pulling same-camera features together forces model to retain domain-specific variation, even in the identity slice.
   - Decrease: Rank-1 (-9.65|-15.10) and mAP (-9.5|-15.75) compared to 1a
2. **Gradient Conflict (1a+1b)**: For any pair (x_i, x_j) where the identity is same but camera is different
   - $\mathcal{L}_{1a}$ treats them as a positive pair (gradient pulls them together); $\mathcal{L}_{1b}$ treats them as a negative pair (gradient pushes them apart).
   - Decrease: Rank-1 (-8.0|-6.35) and mAP (-7.5|-6.5) compared to 1a
3. **Gradient Conflict (1a+1b w/ Ldom)**: Domain loss uniformly distributes (locally) the features of each dataset, whereas 1b promotes camera-based clustering (subset of dataset label)
   - Increases: Rank-1 (+5.9|+7.3) and mAP (+3.7|+7.35) compared to w/o Ldom
4. **Best Performer (1a + 1c)**: Yields observed only in domain conditioning + Finsler domain loss
   - Increases: Rank-1 (+0.6) and mAP (+1.2) compared to 1a
5. **Unified Finsler :** With Finslerized Domain Loss, Unified Finsler formulation outperforms/matches the baseline. With Euclidean Domain Loss, it falls matches/falls shortly behind (~0.5)
6. **Finsler vs. Euclidean Domain Repulsion**: Exact contribution of Finsler Domain Repulsion unclear vs. RNG

Naïve Conclusion
When domain-conditioning the drift vectors, they should undergo camera-based repulsion rather than clustering for generalizability, but identity-based clustering (1, 2, 3, 4)

1. It does so without requiring an explicit Euclidean Identity Triplet slice.
   - *Hypothesis*: The asymmetric Finsler distance, with a regularized multiplier omega, naturally resolves identity and domain slice distinction and domain-shift magnitude/direction on its own, rendering rigid subspace partitioning redundant.
2. Decreases indicates a collapse in generalizability
   - *Hypothesis:* Destroys the identity manifold into domain-specific sub-manifolds, directly violating the primary objective of extracting *domain-invariant* identity features.
3. 1a promotes identity-based clustering, while 1b promotes camera-based clustering
   - *Hypothesis*: Gradient interference stalls the network
   - Destructive interference (instead of regularizing) theoretically justifies the orthogonal decoupling of identity and drift vector slice in the drift head

4. Gradient interference helps regularize the network
   - Disabling L_dom causes a severe performance collapse when using

domain conditioning, regardless of whether the metric space is Euclidean or Finslerized.

- Drop is significantly steeper in the Euclidean space (-13.4 Rank-1 vs -10.0 Rank-1).
- When using instance conditioning, removing L_dom produces marginal shifts,
    - **Conclusion**: Domain-conditioned drift fundamentally requires BAU domain repulsion (L_dom) to prevent feature collapse, whereas instance-conditioned drift can survive without it; presumably as regularizer to prevent degeneration of the drift head

1. d

[Model] Asymmetric Ranking during Evaluation

If `--eval-drift true` is going to improve performance, the drift vector must have received meaningful gradient signals to encode non-commutative domain shifts without collapsing to zero, and the core identity features must not be shattered.

[Results] Best Model Configuration vs. Protocol-2

	M+MS+CS → C3		M+CS+C3 → MS		MS+CS+C3 → M		Average	
	mAP	Rank-1	mAP	Rank-1	mAP	Rank-1	mAP	Rank-1
CLIP-FGDI	**44.4**	44.6	**31.1**	**59.4**	**79.4**	**91.3**	**51.6**	**65.1**
ReNorm	43.6	44.7	25.6	55.6	72.7	89.1	47.3	63.1
ReNorm*	42.4	41.5	24.2	53.3	72.6	88.9	46.4	61.2
BAU	42.8	43.9	24.3	50.9	77.1	90.4	48.1	61.7
BAU*	42.21	42.5	23.28	49.70	76.61	89.61	47.37	60.60
Finsler BAU[1]	43.95	43.93	22.22	48.04	76.47	89.91	47.55	60.63
Finsler BAU[2]	44.27	44.57	22.79	49.16	76.26	89.28	47.77	61.00

[Theory] What can we conclude so far?

1. Increase in model capacity

[Administrative] Requirements of the report

1. **Length**: 8-12 pages in English
2. **Submission deadline**: Max. 6 months (17.04.2026), directly to the supervisor

Is there a presentation?

Generalized Person Re-Identification with Finsler Manifolds
Guided Research (Master Robotics, Cognition, Intelligence Master, 16)

Chair	Informatik 9 - Lehrstuhl für Computer Vision and Artificial Intelligence
Supervisor	Thomas Dages, Simon Weber
Examiner(s)	Daniel Cremers
Begin of thesis	17.10.2025
Due Date	17.04.2026 Add to Calendar
Accepted	Yes
Action	Upload the project Request Extension

Next Week

1. Retry 1b but detach the identity slice such that backpropagation on this loss term only influences the drift vector

2. Read up on feature disentanglement in this particular domain – what else am I missing?
3. Continue writing the report

Prove: that Finsler asymmetry captures domain drift without corrupting ID invariance,

1. **Toy Dataset Construction (Sanity Check):** Generate a synthetic dataset with controlled, non-commutative corruptions (e.g., progressive Gaussian blur, illumination down-scaling) applied to identical PIDs.
2. **Disentanglement Validation:** Extract embeddings from the Toy Dataset. Compute the variance of the 2048-d ID vector and the N-d drift vector. The ID vector variance must approach zero across corruptions, while the drift vector magnitude must correlate monotonically with corruption severity.
3. **Asymmetric Evaluation Activation:** Execute parallel sweeps on Protocol-2 utilizing both --eval-drift false and --eval-drift true. Quantify the delta in mAP. If Finsler asymmetry functions correctly, --eval-drift true will yield superior performance on unseen target domains by explicitly accounting for the source-to-target domain shift during the ranking phase.

When configuring the evaluation with --eval-drift true, how does the specific Finsler distance function utilized in this repository mathematically compute the asymmetric distance between a source query and an unseen target gallery?
For the planned Toy Dataset, what specific synthetic corruptions are you selecting to model non-commutative visual domain gaps that strictly necessitate an asymmetric distance metric over a symmetric one?
In the absence of cross-dataset identities in Protocol-2, how are you ensuring that the drift vector learns a generalized representation of "domain shift" rather than simply memorizing the specific camera parameters of the source training sets?